

Strong Simulation of 1D Quantum Circuits via Reduced Transition Matrices

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Tensor networks are powerful tools for simulating quantum many-body systems, but the growth of spatial entanglement severely limits the direct evolution of pure states under chaotic unitary circuits. Here we consider a more targeted task: the approximate strong simulation of a specified output probability of a 1D chaotic brick-wall circuit at finite relative precision. Given input and output bit strings $\mathbf{y}$ and $\mathbf{x}$, we evaluate $p(\mathbf{x}|\mathbf{y}) = |\langle \mathbf{x} | U(T) | \mathbf{y} \rangle|^2$ using the Sweeping RTM algorithm, a transverse tensor-network contraction based on reduced transition matrices (RTMs). The algorithm compresses the left and right temporal boundary states jointly, seeking an output probability that converges across spatial cuts and as the bond dimension is increased. For chaotic one-dimensional brick-wall circuits at a fixed relative target precision, we find numerical evidence that the bond dimension required to obtain stable estimates grows subexponentially over the accessible time window. Our findings open a direct route to classical probability queries for chaotic quantum circuits, with potential applications to benchmarking and learning tasks.

Introduction. The simulation of nonequilibrium quantum many-body states is limited by the rapid growth of entanglement. Tensor-network methods based on matrix product states (MPS), including time-evolving block decimation (TEBD), time-dependent density matrix renormalization group (tDMRG), and the time-dependent variational principle (TDVP), mitigate this difficulty by restricting the dynamics to low-complexity variational manifolds. These approaches build on density matrix renormalization [1–3], adaptive real-time evolution [4–8], and subsequent developments [9–11]. Tensor-network methods have also proved powerful tools for benchmarking quantum processors, even outperforming current devices in accuracy and, in some cases, speed on specific many-body simulation tasks [12–15]. Their computational cost is controlled by the spatial entanglement of the evolving state, and they therefore eventually become inefficient when this entanglement grows extensively [16, 17]. In many applications, however, one does not need to represent the full time-evolved state. Here we consider the more targeted problem of evaluating the probability of a specified measurement outcome,

$$p(\mathbf{x}|\mathbf{y}) = |\mathcal{A}_{\mathbf{x}\mathbf{y}}(T)|^2, \quad \mathcal{A}_{\mathbf{x}\mathbf{y}}(T) = \langle \mathbf{x} | U(T) | \mathbf{y} \rangle, \quad (1)$$

to fixed *relative* precision ϵ_p (or equivalently a fixed *relative* precision ϵ_A on the amplitude $\mathcal{A}$).

This probability-query task is distinct from sampling the circuit’s output distribution. Sampling from sufficiently deep chaotic circuits is conjectured to be classically hard and forms the basis of quantum-advantage experiments [18–20]. In suitable local random-circuit ensembles, anticoncentration occurs at depth of order

$\log N$, with typical probabilities of order 2^{-N} for a system of N qubits [21]. A fixed additive error is therefore insufficient to resolve these probabilities. Estimating them to fixed relative precision through repeated measurements also requires exponentially many samples.

Computing specified output probabilities is generically hard: exact evaluation is $\#P$ -hard even on average for suitable random-circuit ensembles [22], and this hardness has been extended to exponentially small additive approximation errors [23, 24]. Importantly, however, the proven approximation regimes correspond, at the typical probability scale $p \sim 2^{-N}$, to relative errors that vanish with system size. Here we instead consider fixed finite relative precision, a regime for which computational hardness has not been established.

For a one-dimensional circuit, a natural way to approach this problem is to represent the amplitude in Eq. (1) directly as a two-dimensional space-time tensor network [25–29]. This space-time network can be treated as a generic tensor-network contraction problem, using optimized paths for exact contraction or generic low-rank compression schemes [30–33]. Contraction along the time direction corresponds to ordinary MPS time evolution and inherits its limitations due to spatial-entanglement growth. One may instead contract the network transversely, i.e. along the spatial direction, introducing temporal boundary states associated with the left and right parts of the circuit [25–27, 34, 35]. A transverse reformulation does not, by itself, remove the computational difficulty, since the temporal boundary states may also develop volume-law entanglement [36, 37].

The central observation is that the quantity relevant

for the contraction is not either temporal state separately, but their overlap. This motivates an overlap-based compression using the reduced transition matrix (RTM) of the two boundary states [28, 34]. Generalized temporal entropies constructed from RTMs can remain small even when the individual boundaries are strongly entangled [38–40]. This suggests that the information needed for the overlap can be represented more compactly [28, 38].

Building on recently introduced frameworks [38, 41], we introduce a *Sweeping RTM algorithm*, which constructs the left and right temporal states together as temporal MPS (tMPS), rather than approximating either boundary independently. It proceeds by sweeping across the spatial direction and truncating their temporal bonds according to the RTM associated with their overlap. At each step, both boundaries are reoptimized, and internal consistency checks are used to determine whether the resulting amplitude has converged within the prescribed tolerance.

Our central numerical observation is that working at finite target precision substantially reduces the amount of temporal information that must be retained. Although the temporal boundary states themselves become strongly entangled, the singular spectra relevant to their overlap develop approximately exponential tails, and the bond dimension required to obtain a stable contraction grows much more slowly than the one required for a faithful representation of either state separately. Over the accessible range of system sizes and times, this growth is compatible with subexponential scaling in the simulation time. We stress, however, that our observations do not rigorously establish an asymptotic complexity bound, but provide heuristic evidence.

We validate the method by comparison with exact results for small systems and, for much bigger sizes ($N = 60$), by examining the convergence of output-probability statistics towards the Porter–Thomas prediction for their log moment. This provides a nontrivial yet indirect test of the validity of our algorithm.

We first introduce the RTM and the sweeping algorithm, then examine the numerical accuracy, resource requirements, and retained singular spectra. We conclude with applications to benchmarking and learning, and open questions.

RTM contraction and spectral complexity. The use of transverse contraction to compute expectation values after quantum quenches was introduced in Ref. [25]. In this approach, the computational complexity is controlled by the temporal entanglement of the boundary tMPS generated by contracting the space-time tensor network along the spatial direction. A key observation of Refs. [28, 34] is that this contraction can be improved by exploiting the fact that the final quantity depends only on the overlap between the left and right temporal boundaries. One can therefore target this overlap directly, rather than approximating each boundary inde-

pendently, see Fig. 1.

This observation transfers naturally to the computation of transition amplitudes. If contractions from the left and right define $\langle L|$ and $|R\rangle$, respectively, the desired amplitude is $\langle L|R\rangle$. The corresponding transition matrix is

$$\mathcal{T} = \frac{|R\rangle\langle L|}{\langle L|R\rangle}, \quad (2)$$

and its reduction across a temporal bipartition is $\mathcal{T}_A = \text{Tr}_{\bar{A}} \mathcal{T}$. Here $\langle L|$ denotes the covector obtained from the left network contraction and is not, in general, the Hermitian conjugate of the right boundary. Consequently, $\mathcal{T}_A$ is generally non-Hermitian.

The Sweeping RTM algorithm uses the singular-value decomposition (SVD) of $\mathcal{T}_A$ to identify the temporal subspaces that contribute most strongly to the overlap. The corresponding left and right singular subspaces define the isometries used to compress the two temporal boundary MPS. The SVD provides a natural local criterion for identifying the directions carrying the largest weight of $\langle L|R\rangle$, rather than relying on the entanglement structure of either boundary separately. The bond dimension χ fixes the number of singular directions retained in this compression.

The singular spectrum of $\mathcal{T}_A$ provides a useful diagnostic of the structure retained by this compression. Denoting its singular values by $\{\lambda_n\}$, we define the RTM SVD entropy using base-two logarithms for convenience,

$$S_{\text{RTM}}(T) = - \sum_n w_n(T) \log_2 w_n(T), \quad (3)$$

$$w_n(T) = \frac{\lambda_n(T)}{\sum_m \lambda_m(T)}.$$

This quantity, also studied in [40], is distinct from the temporal entanglement entropy of either boundary separately [28, 39], and is related to the generalized temporal entropies studied in Refs. [38, 42].

Previous studies found bounded or logarithmically growing generalized temporal entropies in continuous-time Hamiltonian dynamics [38, 42], even in regimes where the temporal entanglement of the individual boundary states suggests a much more costly contraction [14]. Recent work has also related truncation errors for local observables to the RTM singular spectrum and established at most logarithmic growth of the corresponding SVD entropy in chaotic random dual-unitary circuits [40]. The latter result, however, concerns only the computation of local observables and does not directly determine the complexity of fixed-output probability calculations. Moreover, the circuits considered here need not admit the continuous-time or conformal-field-theory descriptions underlying some of the earlier results.

Sweeping RTM algorithm. We now describe the Sweeping RTM algorithm in detail, following the sequence illustrated in Fig. 1. For fixed input and output

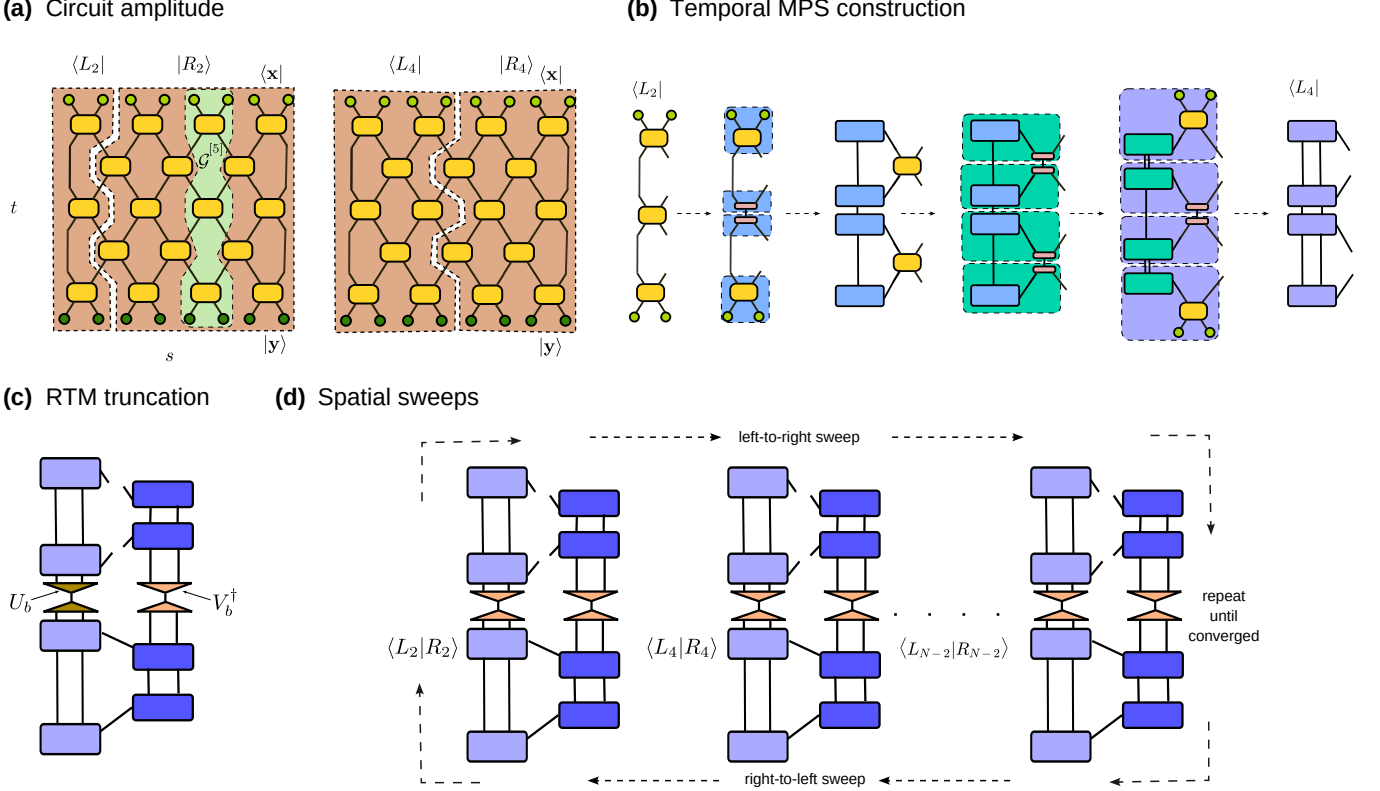


FIG. 1. **Tensor-network representation of the amplitude $\mathcal{A}_{\mathbf{x}\mathbf{y}}$ and its approximate contraction.** (a) $\mathcal{A}_{\mathbf{x}\mathbf{y}}$ is the matrix element of the brick-wall circuit generated by the yellow gates between the product states $|\mathbf{y}\rangle$ and $\langle\mathbf{x}|$ (green dots), with time running upward. At any spatial cut, the spatio-temporal TN can be viewed as the overlap $\langle L_j|R_j\rangle$ of left and right temporal states. Two examples, for $j = 2$ and $j = 4$, are highlighted by the brown shaded regions. These temporal states can be represented as tMPS, whose exact bond dimension can grow exponentially with the distance from the spatial boundary as an increasing number of tensor columns $\mathcal{G}^{[j]}$ (shaded in pale green) are absorbed, before reaching a finite-size or finite-time ceiling. (b) Starting from the left boundary, we illustrate the elementary tensor decompositions that bring $\langle L_2|$ into tMPS form. Absorbing the next two columns, $\mathcal{G}^{[3]}$ and $\mathcal{G}^{[4]}$, produces $\langle L_4|$, which can again be written as a tMPS with enlarged temporal bonds. Repeating this procedure without compression can therefore produce exponential bond-dimension growth. (c) To obtain an efficient approximate contraction, after each absorption step we truncate the tMPS back to bond dimension χ . Each bond b is projected onto the dominant singular subspaces of the corresponding reduced transition matrix (RTM), obtained from the partial contraction of $\langle L_j|R_j\rangle$. The matrices U_b and $V_b^\dagger$ contain the corresponding left and right singular vectors, as in Eq. (8). The sketch shows the RTM associated with the cut at time $t = 2$ for an evolution of total duration $T = 4$. (d) Starting from randomly initialized temporal states $\{\langle L_j|, |R_j\rangle\}$, we iteratively optimize all spatial bipartitions by alternating left-to-right and right-to-left sweeps. At each cut j , the corresponding RTMs are recomputed and the tMPS bonds are truncated as in (c). Sweeps are repeated until the results converge within the prescribed tolerance. If convergence is not reached after a fixed number of sweeps, the bond dimension χ is increased and the procedure is restarted.

bit strings $\mathbf{y}$ and $\mathbf{x}$, we first absorb the physical boundary indices into the circuit gates. The resulting space-time tensor network can then be viewed as a contraction of spatial columns of tensors. We denote the column at site j by $\mathcal{G}_{x_j y_j}^{[j]}$ (for brevity in the following we'll omit the x and y configuration indices); representative columns are shaded in pale green in Fig. 1(a). The amplitude can be written schematically as

$$\mathcal{A}_{\mathbf{x}\mathbf{y}}(T) = \langle L_2 | \mathcal{G}^{[3]} \mathcal{G}^{[4]} \dots \mathcal{G}^{[N-3]} | R_{N-2} \rangle, \quad (4)$$

where $\langle L_2 |$ and $| R_{N-2} \rangle$ contain the contractions at the two outer spatial boundaries. Equation (4) is a com-

pact notation: in the implementation we do not explicitly construct $\mathcal{G}^{[j]}$ as a matrix product operator (MPO), but instead apply its constituent transverse two-site gates directly.

At any spatial cut between columns j and $j + 1$, the circuit can be divided into a left and a right partial contraction,

$$\langle L_j | = \langle L_2 | \prod_{m \leq j} \mathcal{G}^{[m]}, \quad | R_j \rangle = \prod_{m > j} \mathcal{G}^{[m]} | R_{N-2} \rangle. \quad (5)$$

These are the two temporal boundary states highlighted in Fig. 1(a) for two representative spatial cuts. Before

any truncation,

$$\mathcal{A}_{\mathbf{xy}}(T) = \langle L_j | R_j \rangle \quad (6)$$

for every cut j . The *exact* overlap is clearly independent of where the network is divided, however constructing $\langle L_j |$ and $| R_j \rangle$ exactly is exponentially costly (see Fig. 1(b)) and truncating the vectors introduce discrepancies between the cuts. The algorithm approximates each $\langle L_j |$ and $| R_j \rangle$ by temporal MPS with maximum bond dimension χ . The two tMPS are not compressed independently: Instead, the truncation is chosen so as to preserve their overlap, as illustrated in Fig. 1(c).

Consider a temporal bond b of a pair $\langle L_j |, | R_j \rangle$. The reduced transition matrix associated with that bond is

$$\tau_b = \frac{\text{Tr}_{\bar{A}_b} | R_j \rangle \langle L_j |}{\langle L_j | R_j \rangle}, \quad (7)$$

where A_b and $\bar{A}_b$ are the temporal subsystems on either side of bond b . Its singular-value decomposition,

$$\tau_b = U_b \text{diag}(\lambda_1, \lambda_2, \dots) V_b^\dagger, \quad (8)$$

identifies the directions most relevant to $\langle L_j | R_j \rangle$, as sketched in Fig. 1(c). The matrices U_b and $V_b^\dagger$ contain the corresponding left and right singular vectors. We retain at most χ dominant singular directions and absorb the resulting transformations into the neighboring tensors of the two tMPS. In this way, information is discarded according to its importance for the overlap rather than for either boundary state separately. The local implementation operates directly on the canonicalized tMPS tensors and never constructs the exponentially large reduced transition matrix explicitly [28, 41].

For a fixed maximum bond dimension χ , we repeat this procedure in sweeps across the system, alternating left-to-right and right-to-left passes, as sketched in Fig. 1(d).

After every update we record the overlap

$$\mathcal{A}_j^{(x)} = \langle L_j^{(x)} | R_j^{(x)} \rangle \quad (9)$$

at the visited spatial cut j . A completed sweep provides values at every j : in the absence of truncation these values would all coincide. Their spread therefore provides a first internal consistency check of the approximate contraction. We define the following *three internal diagnostics*,

$$\begin{aligned} \Delta_{\text{cut}} &= \max_{j,k=1}^N \frac{|\mathcal{A}_j^{(x)} - \mathcal{A}_k^{(x)}|}{|(\mathcal{A}_j^{(x)} + \mathcal{A}_k^{(x)})/2|}, \\ \Delta_{\text{sweep}} &= \max_{j,k=1}^N \frac{|\mathcal{A}_j^{(x)}(\text{prev}) - \mathcal{A}_k^{(x)}|}{|(\mathcal{A}_j^{(x)}(\text{prev}) + \mathcal{A}_k^{(x)})/2|}, \\ \Delta_\chi &= \max_{j,k=1}^N \frac{|\mathcal{A}_j^{(x')} - \mathcal{A}_k^{(x)}|}{|(\mathcal{A}_j^{(x')} + \mathcal{A}_k^{(x)})/2|}, \end{aligned} \quad (10)$$

where *prev* refers to the previous sweep and $\chi' > \chi$ is the next allowed bond dimension value (we choose trial bond dimensions from the grid $\chi = 16 + n^2$ and allow a number of sweeps of order $\sim N/2$ at each step. The three diagnostics must remain below the internal relative tolerance $\epsilon_{\mathcal{A}}$. Notice that one can also fix the error on the probabilities $p = |\mathcal{A}|^2$ by changing $\mathcal{A}$ with their square modulus and taking ϵ_p as threshold. Here for Fig. 2 and 3, we choose a relative tolerance on the amplitude $\mathcal{A}$, denoted by $\epsilon_{\mathcal{A}}$. A candidate amplitude satisfying these criteria is subjected to a even further validation stage at a larger bond dimension $\chi' > \chi$. This series of checks allows us to make sure that the sweep is not stuck in local minima of the internal diagnostic. Once the checks pass, we report

$$p(\mathbf{x}|\mathbf{y}) = |\mathcal{A}_{N/2}^{(x)}|^2 \quad (11)$$

together with its maximum retained bond dimension χ at which convergence was reached. Empirically, too few sweeps can lead to an unnecessarily large bond dimension because the optimization has not reached its fixed point.

We emphasize that the internal convergence checks above do not provide rigorous error bounds: we do not prove that satisfying them guarantees a prescribed relative error in the amplitude, nor that the sweep optimization always converges. For the cases studied here, we find numerically that the error remains below the prescribed threshold. A rigorous justification and, if possible, a well-defined variational formulation would nevertheless be desirable.

Results. We benchmark three brick-wall families of two-qubit gates with open spatial boundaries. The system size is denoted by N , while time T counts complete odd-even periods. The deterministic benchmark is a chaotic kicked-Ising-type Floquet circuit, inspired by the standard kicked Ising setting [27]. On each bond we apply

$$U_{\text{KI}} = \exp[-iH_X] \exp[-iH_Z], \quad (12)$$

with

$$\begin{aligned} H_X &= X \otimes X + 0.81 \frac{X \otimes I + I \otimes X}{2}, \\ H_Z &= 0.904508 \frac{Z \otimes I + I \otimes Z}{2}. \end{aligned} \quad (13)$$

Here X and Z are Pauli matrices and I is the identity. The same gate is repeated on all active bonds and at all periods. For the exchange-symmetric random ensemble, we draw U_0 from Haar measure on $U(4)$, form $M = (U_0 + P_{\text{sw}} U_0 P_{\text{sw}})/2$, and use its polar unitary $U_{\text{sym}} = VW^\dagger$, where $M = V\Sigma W^\dagger$ and P_{sw} swaps the two qubits. Thus $[U_{\text{sym}}, P_{\text{sw}}] = 0$. The random Haar ensemble uses U_0 without symmetrization. Random gates are drawn independently for each period, layer, and bond.

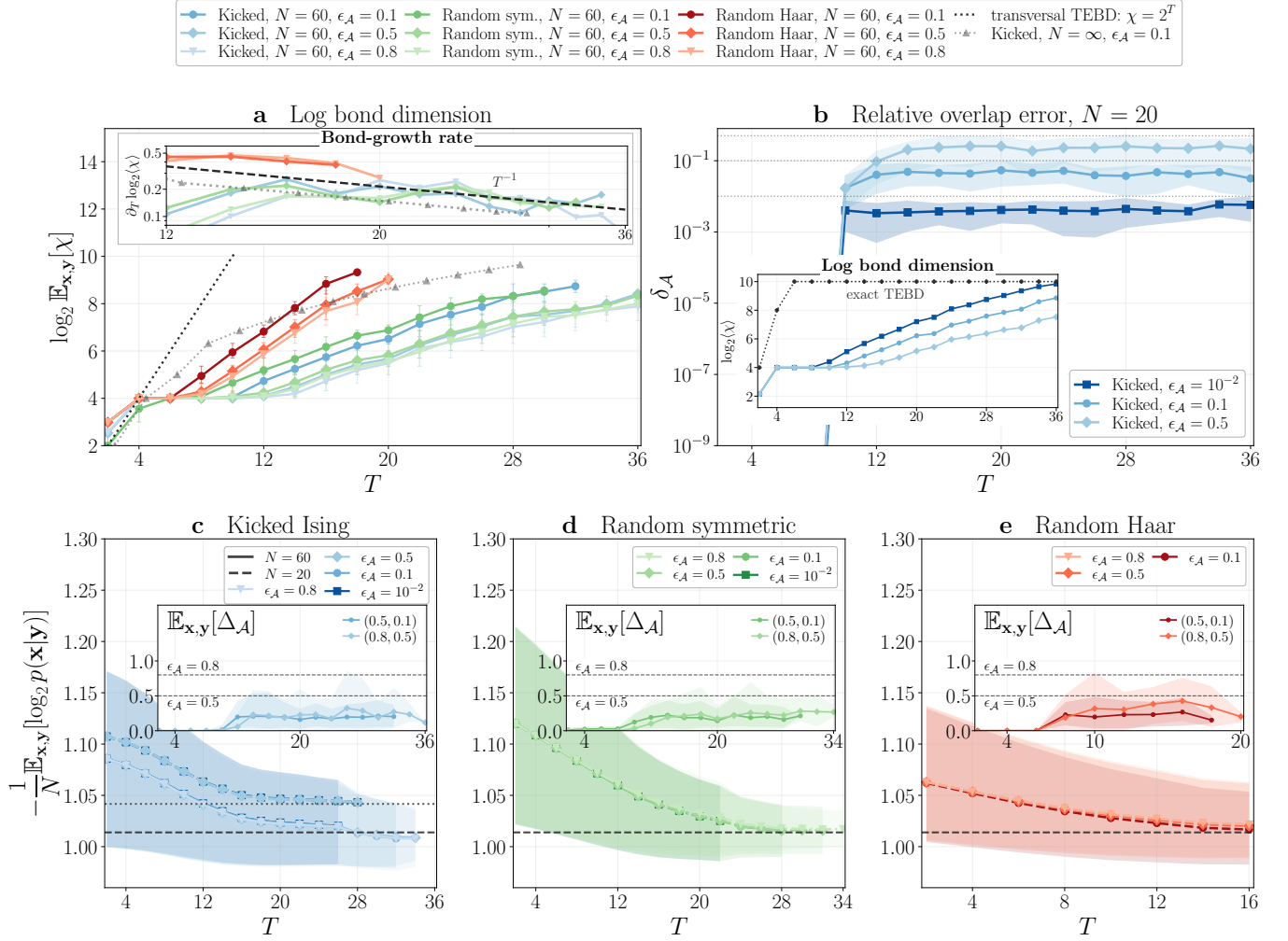


FIG. 2. **Sweeping RTM algorithm: strong-simulation diagnostics.** (a) Logarithm of the mean bond dimension at $N = 60$, $\epsilon_A = 0.8, 0.5, 0.1$ averaged over different bitstrings. The transversal TEBD corresponds to standard TEBD evolution of the left or the right temporal vectors. The inset shows the logarithmic bond-growth rate, with a T^{-1} visual guide (not a fit) which would correspond to a polynomial growth of χ . Gray triangles show the translationally invariant kicked-Ising $N = \infty$, for the configuration $\mathbf{x} = \mathbf{y} = \mathbf{0}$ calculation at $\epsilon_A = 0.1$. (b) Kicked-Ising relative amplitude error at $N = 20$ against exact forward evolution, with the shading representing the different bitstrings values. The inset shows the corresponding log mean bond dimension, as well as the required one for an exact standard TEBD computation. (c)–(e) Mean intensive log probability for kicked Ising, random symmetric, and random Haar circuits at the tolerances shown. Data are smoothed in time using a Gaussian kernel with standard deviation $\sigma_T = 5$. Panel (c) also includes $N = 20$ without shading; the other curves have $N = 60$. Horizontal references include the finite-size Haar shift eq. (14). Insets in (c)–(e) show $\mathbb{E}_{x,y}[\Delta_A]$, where $\Delta_A = |\hat{A}_{\epsilon_A} - \hat{A}_{\epsilon'_A}| / |\hat{A}_{\epsilon'_A}|$. Legend pairs denote $(\epsilon_A, \epsilon'_A) = (0.5, 0.1)$ and $(0.8, 0.5)$. Solid curves have $N = 60$; the dotted curve in (c) has $N = 20$. Dashed horizontal lines mark $\epsilon_A = 0.5, 0.8$. In all figures shading represents the interval between 5th–95th percentiles.

Figure 2 summarizes the strong-simulation diagnostics for the kicked-Ising, exchange-symmetric random, and random Haar circuits. We evaluate independently chosen, uniformly distributed computational-basis bitstring pairs $(\mathbf{y}, \mathbf{x})$, keeping the circuit realization and the sampled pairs fixed across times. We average over up to 40 bitstring pairs, with the number depending on time and circuit, using a single circuit realizations of each circuit family. Panel (a) shows the growth of the averaged

bond dimension for $N = 60$ and $\epsilon_A = 0.8, 0.5, 0.1$. A fixed exponential growth of the required bond dimension would appear as a straight line on these axes. Instead, after an initial transient, the curves develop a clear downward curvature, most visibly for the kicked-Ising and exchange-symmetric random circuits. The inset quantifies this slowdown through the logarithmic bond-growth rate. The T^{-1} guide would correspond to polynomial growth of χ . The data support subexponential growth

over the measured window, with a more limited comparison available for random Haar circuits where the computations are substantially harder. The dotted 4^T reference indicates the maximal bond-dimension growth of direct forward evolution.

For comparison, we also report the translational invariant case $\mathbf{x} = \mathbf{y} = \mathbf{0}$ contraction for an infinite ($N = \infty$), translationally invariant kicked-Ising circuit. In this case, we use a power method rather than sweeps, with convergence determined by a relative tolerance, as in the sweeping algorithm. Similarly to the CFT cases in Refs. [38, 42], this case shows the clearest logarithmic growth, with a clear T^{-1} decay of the logarithm of the bond-dimension growth, unlike the finite-size simulations, which show a faster growth at intermediate times (see also next section).

In panel (b), we validate the internal error control against amplitudes obtained by exactly evolving the initial state and projecting the evolved state onto the desired output bit string. We perform this comparison for a small system of $N = 20$, for which exact evolution is feasible. The mean relative amplitude error remains below the corresponding internal tolerance over the accessible time window.

To test the strength of the method, however, we need to go beyond small sizes. A direct comparison with exact results for $N = 60$ is not feasible. In panels (c)–(e), we provide an independent statistical consistency check in the regime where exact forward evolution is unavailable. We evaluate the mean intensive log probability for uniformly queried bit-string pairs, not Born-distributed samples. The finite-size prediction for a complex Haar-random state is

$$\mathbb{E}_{\mathbf{x}, \mathbf{y}} \left[-\frac{1}{N} \log_2 p(\mathbf{x}|\mathbf{y}) \right] = 1 + \frac{\gamma_E \log_2(e)}{N} + O\left(\frac{2^{-N}}{N}\right), \quad (14)$$

where γ_E is Euler’s constant [19, 43]. Panel (c) shows the finite-size references for $N = 60$ and $N = 20$, while panels (d) and (e) use $N = 60$. In all three ensembles the data approach the corresponding Haar references at later times, although with some strong oscillations, possibly due to the single circuit realization and the finite number of bitstrings considered. For this reason, we present time-averaged data in the plot. The error given by the finite threshold is expected to be $\epsilon_p/(N \ln 2)$, which is in agreement with the curves displayed in Fig. 2; the near coincidence of the curves, despite their different retained bond dimensions, is therefore consistent with controlled relative errors and supports the stability of the algorithm across tolerances. Moreover, in the insets of Fig. 2 (c)–(e), we also display the relative discrepancy between the amplitudes computed at different thresholds ϵ_A showing that also the latter remains bounded.

RTM spectra and SVD entropy. We next examine the retained RTM spectra in Fig. 3. Panels (a) and

(b) show representative singular spectra for the kicked-Ising and exchange-symmetric random circuits. Over the accessible time window, their resolved tails are approximately described by

$$\frac{\lambda_n(T)}{\lambda_1(T)} \simeq C(T)e^{-n/F(T)}, \quad (15)$$

where $F(T)$ is the characteristic decay scale, extracted from a linear fit to the logarithm of the tail. Each displayed spectrum uses one bit-string pair and one central spatial and temporal cut.

Panel (c) summarizes the corresponding decay scale averaged over the sampled input-output pairs. Over the accessible time window, its mean is approximately described by $\mathbb{E}_{\mathbf{x}, \mathbf{y}}[F(T)] \simeq BT^\alpha$. An exactly exponential singular spectrum provides a simple mechanism for logarithmic SVD-entropy growth. If the full normalized spectrum were geometric, $w_n = (1 - q)q^{n-1}$ with $q = e^{-1/F}$, then for $F \gg 1$

$$S_{\text{RTM}} = \log_2 F + O(1). \quad (16)$$

A characteristic scale $F \propto T^\alpha$ would consequently imply $S_{\text{RTM}} = \alpha \log_2 T + O(1)$. The spectra in panels (a,b), however, have additional structure among their largest singular values, outside the fitted tail. The evolving weight of these values can produce deviations from the logarithmic behavior suggested by the fitted tail alone. Consistently, the finite-size entropy in panel (d), averaged over the maximum local entropy of each run, does not yet display a clean logarithmic law.

By contrast, the translationally invariant $N = \infty$ kicked-Ising calculation for $\mathbf{0} \rightarrow \mathbf{0}$ boundaries is much more compatible with approximately logarithmic growth over the displayed window. As with the bond-dimension results in the previous section, this closely recalls the logarithmic generalized temporal entropy found in critical continuous-time dynamics. There is therefore a clear distinction between the infinite, translationally invariant case and the finite-size case with generic bit strings, in terms of complexity and RTM-entropy scaling. This can be understood in terms of the infinite-size limit projecting onto the stationary left and right states of the transfer matrix obtained by the product of two consecutive $\mathcal{G}$ s, which possibly have low complexity, while the finite-size case also involves its excited states. Characterizing these states, as well as the stationary state in the absence of CFT symmetry, remains a challenging problem for future work.

It is also interesting to note that finite-precision evaluation does not reduce the entanglement of the exact evolved state; rather, it allows information that is not required to reproduce the scalar overlap at the targeted precision to be discarded. For a retained RTM spectrum of rank χ ,

$$S_{\text{RTM}} \leq \log_2 \chi. \quad (17)$$

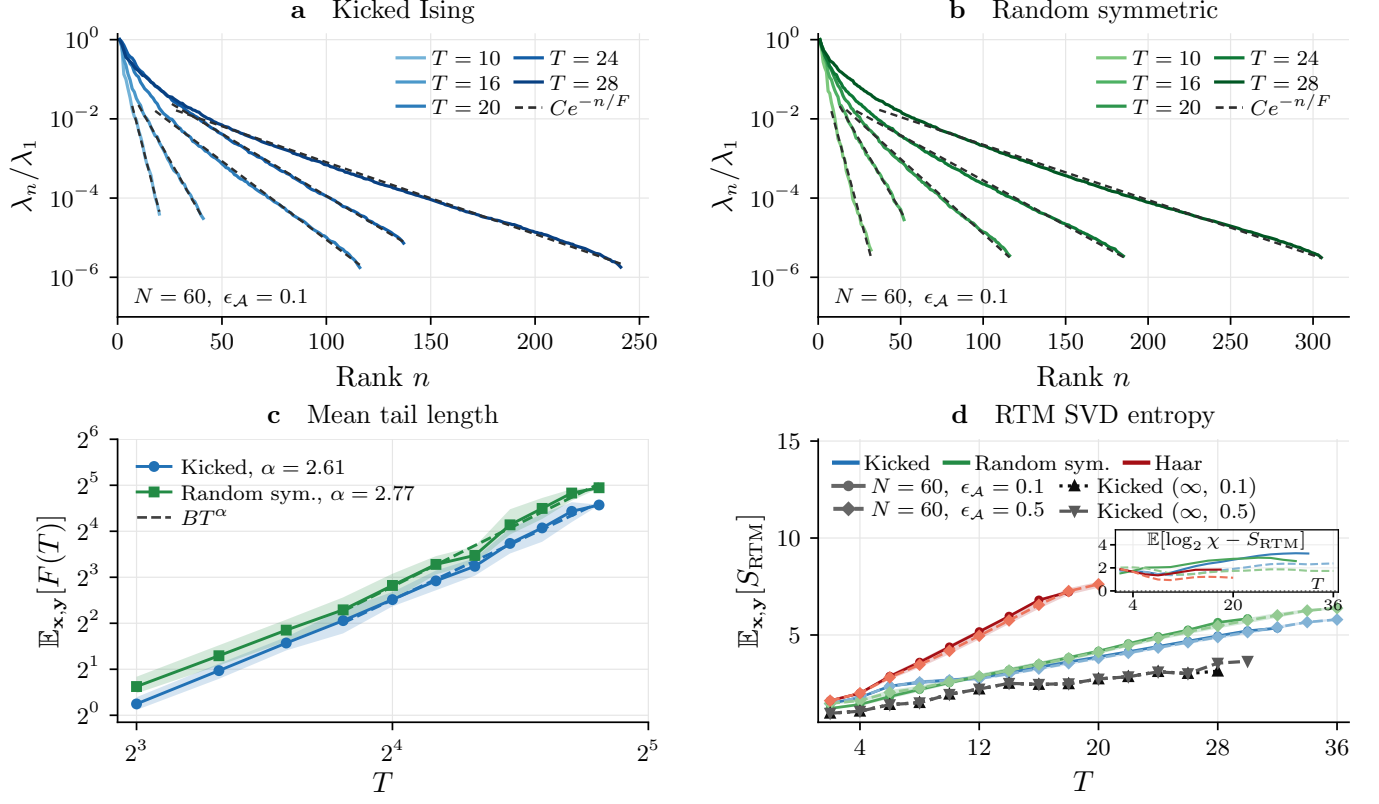


FIG. 3. **RTM spectra and entropy.** (a,b) Single-cut spectra λ_n/λ_1 for kicked Ising and random symmetric circuits at $N = 60$, $\epsilon_A = 0.1$, at different times. Darker shades indicate later times; dashed segments fit the resolved exponential tails. (c) Mean decay length in both models averaged over bitstring with the power-law fits. (d) Mean maximum local RTM SVD entropy for the three $N = 60$ circuit ensembles at $\epsilon_A = 0.1, 0.5$. Black/gray triangles show translationally invariant kicked-Ising $N = \infty$, $\mathbf{x} = \mathbf{y} = \mathbf{0}$ results at $\epsilon_A = 0.1, 0.5$. The inset shows the entropy saturation $\mathbb{E}_{\mathbf{x}, \mathbf{y}}[\log_2 \chi - S_{\text{RTM}}]$ for the finite-size data, with the same colors and line styles.

To test proximity to this ceiling, we evaluate $g = \log_2 \chi - S_{\text{RTM}}$ separately for each run and then average over pairs, as shown in the inset of Fig. 3(d). At later times the finite-chain mean gaps remain of order one to three, rather than approaching $g = 0$. Equivalently, their entropy effective rank $2^{S_{\text{RTM}}}$ is smaller than χ by a factor of order two to ten.

Benchmarking large quantum circuits. A natural application of relative-accuracy probability queries is the classical evaluation of linear cross-entropy benchmarking (XEB) for large quantum devices, without constructing or storing their full ideal output states [19, 20]. For a fixed input $\mathbf{y}$, suppose that the device produces M measured strings $\mathbf{x}_i$. The classical calculation then requires only the ideal probabilities $p(\mathbf{x}_i|\mathbf{y})$ of these sampled strings, rather than all 2^N output probabilities. On this sample, the linear XEB estimator is

$$X = \frac{2^N}{M} \sum_{i=1}^M p(\mathbf{x}_i|\mathbf{y}) - 1. \quad (18)$$

For sufficiently deep chaotic circuits in the weak-noise regime where XEB tracks fidelity, $X \simeq \mathcal{F} = \langle \psi | \rho | \psi \rangle$,

where $|\psi\rangle = U(T)|\mathbf{y}\rangle$ and ρ is the experimental state [20, 44, 45]. Beyond this regime, the depth dependence of XEB can still provide access to fidelity through the finite-depth scaling theory of Ref. [45]. Replacing p by the RTM estimate $\hat{p}$ defines $\hat{X}$ on the same device samples. If each query has relative error at most ϵ_p ,

$$|\hat{X} - X| \leq \epsilon_p(X + 1). \quad (19)$$

Thus, in the regime where $X \simeq \mathcal{F}$, RTM probability queries determine the device fidelity up to classical corrections $\pm \epsilon_p(\mathcal{F} + 1)$, in addition to measurement uncertainty.

A complementary application is the shadow-overlap certification protocol of Ref. [46]. It combines single-qubit measurements with queries of two complex target-state amplitudes. For fixed input $\mathbf{y}$, let $\mathbf{x}^{(0)}$ and $\mathbf{x}^{(1)}$ agree with the measured outcomes on all but qubit i . For a nonzero denominator, the queries specify

$$|\phi_i\rangle = \frac{\mathcal{A}_{\mathbf{x}^{(0)}\mathbf{y}}|0\rangle + \mathcal{A}_{\mathbf{x}^{(1)}\mathbf{y}}|1\rangle}{\sqrt{|\mathcal{A}_{\mathbf{x}^{(0)}\mathbf{y}}|^2 + |\mathcal{A}_{\mathbf{x}^{(1)}\mathbf{y}}|^2}}. \quad (20)$$

The two complex amplitudes determine both the relative weight and phase of this conditional single-qubit state. Its overlap with the measured single-qubit state is estimated from randomized Pauli measurements and averaged to obtain the shadow overlap ω . This gives the fidelity bound $\mathcal{F} \geq 1 - \tau(1 - \omega)$, where τ is the relaxation time of the target-state Markov chain [46]. If each RTM amplitude has true relative error $\delta_{\mathcal{A}} \leq \epsilon_{\mathcal{A}} < 1$, the conditional-state projector changes by at most $\epsilon_{\mathcal{A}}$ in operator norm. The bias in the mean shadow overlap is therefore at most $\epsilon_{\mathcal{A}}$, and the fidelity lower bound is reduced by at most $\tau\epsilon_{\mathcal{A}}$, in addition to the statistical uncertainty. RTM contraction thus supplies the phase-sensitive queries needed to bound the device fidelity, with an error set by the achieved amplitude accuracy and τ .

Finally, probability queries also enable likelihood-based learning of parametrized quantum circuit models [47]. For training strings $\mathbf{x}_i$ and fixed input $\mathbf{y}$, the empirical negative log likelihood is

$$\mathcal{L}(\theta) = -\frac{1}{M} \sum_{i=1}^M \log_2 p_{\theta}(\mathbf{x}_i|\mathbf{y}). \quad (21)$$

If every queried probability has true relative error at most $\epsilon_p < 1$, then $|\hat{\mathcal{L}} - \mathcal{L}| \leq -\log_2(1 - \epsilon_p)$. Thus relative accuracy controls the error in the training objective even when the individual probabilities are exponentially small. This opens a route to evaluating likelihoods without explicitly representing the full output distribution, provided the contractions remain tractable across the parameter region explored during learning.

Complex amplitude queries could additionally supply reference data for neural-network or tensor-network state models, complementing measurement-based tomography [48]. The shadow-overlap framework also supplies objectives for learning state representations and optimizing state preparation [46].

Outlook. Several questions remain open. The present evidence is numerical and heuristic, therefore rigorous bounds need to be found. In particular, it remains to be fully established whether the growth of the bond dimension at a given threshold ϵ_p , i.e. $\chi(T, \epsilon_p, N)$, is exponential, subexponential, or purely polynomial and if the internal checks for the error provided here are enough. A variational approach to the problem of the overlap, similarly to what is done in TDVP evolution [8, 49] would be desirable.

An extension to two spatial dimensions is also under development. In that case the left and right temporal boundaries become projected entangled-pair states (PEPS). The principle of targeting their overlap through a reduced transition matrix continues to apply, but representing, contracting, and compressing the two-dimensional temporal boundaries introduces additional approximations and computational costs [50, 51]. Extending the present strategy is therefore possible in prin-

ciple but nontrivial, and efficient sweeps with verified relative accuracy remain a challenging task.

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